

University as a Melting Pot: Long-Term Effects of Internationalization[†]

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This paper provides the first evidence on the impact of exposure to international students on the long-term outcomes of native students. I combine unique survey and administrative data from the Netherlands covering one million students across three decades and employ an across-cohort design. I find that exposure to international students leads natives to (i) form social ties with nonnatives, (ii) hold more positive attitudes towards migration and learning about other cultures, and (iii) seek opportunities abroad. Notably, I find precisely estimated zero effects on employment, income, entrepreneurship, and the share of international coworkers up to 25 years after university entry. (JEL F22, I23, I26, J15, Z13)

Over the past two decades, the number of students studying abroad has tripled globally (UNESCO 2024). Universities attract international students to increase revenue, expand social networks, and offer students opportunities to work with diverse peers in an international setting. However, politicians have raised concerns about the displacement of domestic students, negative labor market impacts, and the costs of subsidizing the education of foreign students. These concerns, along with broader debates on migration, have led to measures limiting student inflows in countries that host half of all international students, such as Australia, Canada, Denmark, Finland, France, Germany, the Netherlands, Norway, Sweden, Switzerland, the United Kingdom, and the United States.¹ While these policies target international students, they may have unintended consequences for native students.

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¹Since 2023 at least nine countries have implemented restrictive policies targeting international students. The Australian and Canadian governments have introduced annual caps on student visas (Ross 2024; Bhardwa 2024). Meanwhile, tuition fees for international students have increased in Finland (Dixon 2024b), Germany (Upton 2023a), Norway (Upton 2023b), and Switzerland (Etias 2024). The Dutch government is currently reducing the number of English-taught bachelor's programs (Dixon 2024a), and the United Kingdom has recently restricted international

This paper examines the impact of exposure to international students during the first year of a bachelor's program on the long-term outcomes of native students in the Netherlands. I employ an across-cohort design, following the approach introduced by Hoxby (2000), which is particularly well suited to the Dutch context, where the absence of grade-based selection and the lack of fixed capacity constraints make the variation in the number of native and international students largely idiosyncratic. Specifically, I exploit cohort-to-cohort variation in the share of international students within university programs over time. My approach allows for student selection *across* programs, as it does not compare students from different programs but rather compares students *within* the same program *across* different cohorts.

The identification strategy relies on the assumption that students do not select into specific cohorts *within* a program based on the presence of international students. This assumption is plausible for two reasons. First, prospective students are generally unaware of the international student composition in their program until they begin their studies. Second, even if students could estimate the approximate composition, it is unlikely that they would alter their enrollment timing to avoid or seek out international peers. To support this assumption, I provide evidence that the share of international students is uncorrelated with the likelihood of native students taking a gap year, a rich set of predetermined characteristics at the student, sibling, and parent levels, peer composition and indicators of program quality.

The results show that exposure to international students increases natives' social ties with nonnatives and their openness to migration. During university, a 10 percentage point increase in the share of international students leads to a 0.12 standard deviation increase in positive attitudes towards the internationalization of education and a 0.07 standard deviation increase in satisfaction with opportunities to learn about other cultures. Fifteen years after entering university, a 10 percentage point increase leads to (i) a 5.9 percent increase in the probability of cohabiting with a nonnative; (ii) a 4.2 percent increase in the probability of marrying a nonnative, and (iii) a 4 percent increase in the probability of emigration. The results withstand a battery of robustness checks, including adjustments for multiple hypothesis testing, time trends, program size, individual and peer characteristics, different sample definitions, definitions of international students, and a placebo test.

To illustrate the size of the effects, consider a policy whereby a university increases the share of international students in each incoming bachelor's program cohort by 10 percentage points, given an average program size of 137 students. On average, such an increase would result in one additional native student per program cohabiting with a nonnative, one additional native emigrating for every three programs, and one additional native marrying a nonnative for every five programs.

Having established positive effects of international exposure on social ties and openness to migration, I then examine whether the presence of international students

bachelor's students from bringing family members (Jack 2023b). The United States has cut funding for STEM research that supports international students and is considering a major travel ban (Knox 2025). Prior to 2023 three other countries enacted similar measures. France (Matthews 2018) and Sweden (Fearn 2011) implemented increases in tuition fees for international students, while Denmark imposed caps on international student enrollments in 2021 but reversed the policy in 2023, acknowledging unintended negative consequences from restricting foreign student inflows (Jack 2023a).

affects native students' labor market outcomes. Policymakers often voice concerns that international students might strain educational resources and reduce educational quality, potentially disadvantaging natives in the labor market. However, I find that exposure to international students shows precisely estimated zero effects on employment, income, entrepreneurship, and the share of foreign-born coworkers in the firms where natives are employed up to 25 years after university entry. These findings suggest that exposure to international students makes natives more culturally open and internationally oriented, without compromising their local economic success.

Social integration effects are strongest in collaborative and diverse environments and absent in more competitive ones. For example, STEM fields and programs with fewer female students, which tend to be more competitive, show no significant effects on social ties between native and international students. In contrast, non-STEM fields and programs with a more balanced sex composition foster stronger social integration. Additionally, social ties between native and international students are observed only in larger programs since they offer broader networks and more opportunities for interaction. Exposure to EEA+ peers enhances social integration, likely due to shared cultural and institutional backgrounds. Interactions with non-EEA+ students, who face greater cultural and linguistic differences, do not significantly enhance social integration but increase satisfaction with opportunities to learn about other cultures.

Theoretically, two main mechanisms could explain the positive effects on social outcomes: meeting opportunities and preference shift. An important reason to distinguish between these mechanisms is that they imply different patterns of general equilibrium effects. Under the meeting opportunities mechanism, the impact of exposure operates only within the university context and does not fundamentally change cultural attitudes. The reach of the policy is therefore narrow and bounded: only students who directly interact form social ties, which fade once students leave the university environment. In contrast, the preference-shift mechanism can generate more durable effects: Natives exposed to diversity at university may remain culturally open long after graduation. From a policy perspective, this distinction is critical: While the meeting opportunities mechanism might yield short-lived and localized improvements, a shift in preferences can lead to broad and lasting changes in societal openness and integration.

Empirically, I find strong evidence supporting the preference-shift mechanism. If the meeting opportunities mechanism were the sole explanation, intercultural relationships would predominantly form among peers within the same university program. However, natives frequently establish intercultural ties extending beyond their immediate academic environment, developing relationships with nonnatives outside the university. Moreover, exposure to international students significantly alters natives' attitudes: It increases favorable views towards the social integration of foreigners and support for European unification, while reducing negative attitudes towards foreigners moving into the neighborhood. These empirical patterns strongly suggest that exposure to international peers induces meaningful shifts in underlying preferences towards migration and cultural diversity, rather than merely providing convenient meeting opportunities for intercultural contact.

This paper expands existing research by providing new evidence in two areas: (i) the impact of internationalization of higher education and (ii) the relevance of the contact hypothesis in the context of university education.

First, I contribute to the small but growing literature on the impact of the internationalization of higher education. While most existing studies focus on academic and short-term labor market outcomes, this paper is the first to examine the impact of international peers on native students' social outcomes and long-term labor market outcomes.² For example, Anelli, Shih, and Williams (2023) and Rakesh (2023) examine the effects of exposure to foreign classmates in introductory first-year math courses at two US universities. Anelli, Shih, and Williams (2023) find that exposure to foreign students decreases the likelihood of native students graduating with a STEM degree and has no significant impact on expected earnings. Similarly, Rakesh (2023) finds that exposure to foreign students lowers the graduation rate of native students. In the United Kingdom, Chevalier, Isphording, and Lissauskaite (2020) find no impact of exposure to non-English speakers on the academic performance or emigration of native students six months after graduation. Likewise, Costas-Fernández, Morando, and Holford (2023) find no effects of exposure on graduation, employment, and earnings six months after graduation in the United Kingdom. To summarize previous findings alongside my own results, I estimate a meta-analytic pooled estimate of 0.0 percent, with a 95 percent confidence interval of $[-0.002, 0.002]$. Taken together, this indicates no impact of exposure to international students on the educational and labor market outcomes of natives. However, this paper demonstrates that social and cultural dimensions are first-order effects of internationalization, underscoring the importance of moving beyond conventional academic and economic metrics when evaluating its impact.

Second, this paper contributes to the literature on the contact hypothesis, which suggests that contact with minority groups can reduce prejudice.³ As universities become more diverse, they offer a valuable setting for shaping natives' preferences through everyday intercultural interaction. Previous studies in higher education have explored the impact of being assigned a roommate of a different race on interracial prejudice and connections in the United States (Boisjoly et al. 2006; Marmaros and Sacerdote 2006; Camargo, Stinebrickner, and Stinebrickner 2010; Carrell, Hoekstra, and West 2019), and South Africa (Corno, La Ferrara, and Burns 2022).⁴

²Related studies have explored crowd-out effects (Borjas 2004; Shen 2016; Machin and Murphy 2017; Shih 2017; Chen 2023) and the resource implications of internationalization (Zhu 2023). A large body of work has used Hoxby-style designs to study peer effects on academic outcomes, for example in the United States (Whitmore 2005; Bifulco, Fletcher, and Ross 2011) and Israel (Lavy and Schlosser 2011; Lavy, Paserman, and Schlosser 2012). More recently, researchers have applied this approach to outcomes beyond academic achievement. These include labor market outcomes in Norway (Black, Devereux, and Salvanes 2013), the United States (Bifulco, Fletcher, Oh, and Ross 2014; Carrell, Hoekstra, and Kuka 2018; Merlino, Steinhardt, and Wren-Lewis 2019, 2024), and Denmark (Brenøe and Zölitz 2020); interethnic family formation and neighborhood and workplace diversity in the United States (Merlino, Steinhardt, and Wren-Lewis 2019, 2024) and Sweden (Holmlund, Lindahl, and Roman 2023); and attitudes towards minority groups in the United States (Merlino, Steinhardt, and Wren-Lewis 2019, 2024).

³The hypothesis posits that cross-cultural interactions can effectively break down stereotypes and prejudices between groups when a set of conditions, such as common goals, intergroup cooperation, support by social and institutional authorities, and equal status, are satisfied (Williams 1947; Allport 1954).

⁴Related work investigates interventions in primary and secondary schools, including Rao (2019) in India, Merlino, Steinhardt, and Wren-Lewis (2019) in the United States, and Alan, Baysan, Gumren, and Kubilay (2021); Boucher et al. (2021) in Turkey.

This study is the first to test the contact hypothesis in a European university context. Moreover, by leveraging variation in group composition rather than relying on roommate assignments, this paper examines a more scalable and policy-relevant form of exposure.

My findings have significant policy implications for current debates on internationalization. Governments in a dozen countries are implementing policies to limit the number of international students. Based on the results of this paper, reducing international student enrollment could weaken social integration, cultural openness, and international opportunities for native students. This is especially relevant as it affects students during their impressionable years, a critical period when attitudes and preferences are most adaptable.

The remainder of the paper is structured as follows. Section I provides an overview of the Dutch higher education system. Section II describes the administrative and survey data, along with the identification strategy. Section III presents the main results, a heterogeneity analysis, robustness checks, and a meta-analysis. Section IV concludes.

I. Institutional Background

Under the Dutch law, students who complete the academic track in secondary education are eligible for admission to any of the 13 research universities without grade-based selection criteria.⁵ This policy ensures that the influx of international students does not displace Dutch students or affect their access to university education. Under the European Union (EU) law, all EU nationals must meet the same criteria.⁶ This creates an ideal setting for my identification strategy, as year-to-year variation in native and international student numbers is unpredictable for both incoming students and university programs.

There are low barriers for international students to enroll, because many programs use English as the primary language of instruction. Moreover, the tuition fees in the Netherlands are not high compared to those in other popular English-speaking destinations, such as Australia, the United Kingdom, and the United States. Faculty salaries are determined by collective labor union agreements, ensuring a consistent level of quality across institutions. This commitment to accessibility and quality has resulted in all universities consistently ranking among the top institutions globally (THE 2023). As a result of these factors, the share of international students in the Netherlands has seen a remarkable increase, rising from a modest 3 percent in the 1990s to over 20 percent by 2019 (see Figure 1).

At Dutch universities, students enroll in specific programs from the first year. The first year consists of compulsory courses, ensuring that students in the same program share a common curriculum and interact regularly in peer groups. In later years students may select elective courses, allowing for more specialization in their

⁵There are a few exceptions in programs like medicine, dentistry, and veterinary (see Leuven, Oosterbeek, and de Wolf (2013) and Ketel, Leuven, Oosterbeek, and van der Klaauw (2016) for more details).

⁶These rules also apply to nationals of Iceland, Liechtenstein, Norway, and Switzerland.

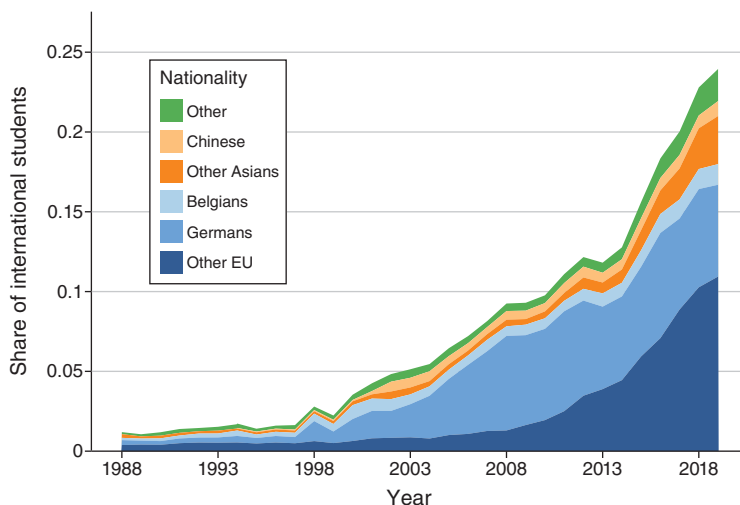


FIGURE 1. THE SHARE OF INTERNATIONAL STUDENTS ENROLLED IN BACHELOR'S PROGRAMS AT DUTCH RESEARCH UNIVERSITIES IN 1988–2019

Note: EEA+ refers to students from the European Economic Area (the EU, Iceland, Liechtenstein, and Norway) plus Switzerland.

Source: Calculated by the author using data from Statistics Netherlands.

field of study. This paper focuses on the first year, where shared compulsory courses provide structured opportunities for peer interaction between native and international students.

Figure 1 shows that around 75 percent of all international first-year bachelor's students originated from other European countries, with German and Belgian students forming the largest groups. Asian students constituted 16 percent of the first-year international student body, with the largest groups being Chinese and Indian.

The share of international students differs substantially across fields of study. In 2019, economics, social sciences, and humanities had the highest shares of international students, around one-third. STEM fields followed with around 20 percent, while law and health had the lowest percentages at around 10 percent. The lower percentages in law and health are primarily due to language barriers, as most programs in these fields are taught in Dutch.

Universities have several incentives to attract international students. In the Netherlands government funding is tied to Dutch and EEA+ student enrollment, making international recruitment financially advantageous. Moreover, non-EEA+ students pay higher tuition fees than their Dutch or EEA+ counterparts, providing additional financial boost. In 2019 tuition fees for Dutch and EEA+ national students were around €2,000 per year, while fees for non-EEA+ nationals were approximately €12,000. Additionally, university rankings frequently consider the share of international students and faculty, enhancing the appeal of a diverse student body.

II. Research Design

In this section I discuss the data sources and my identification strategy.

A. Data

Administrative Data.—I utilize nationwide administrative data provided by Statistics Netherlands (CBS), which cover higher education enrollment from 1988 to 2023, secondary education enrollment from 2006 to 2023, tax registry from 2003 to 2023, firm registry from 2006 to 2023, welfare registry from 2003 to 2023, cohabitation registry from 1994 to 2023, and marriage registry from 1994 to 2023 Statistics Netherlands (CBS) (2024a–l). All registries are linked using unique identifiers. The education registry includes details on secondary school, exam grades, university, program, study duration, dropout, and graduation. The tax registry provides information on employment, entrepreneurship, and earnings. The firm registry contains all firms and their workers. The welfare registry provides information on unemployment and social benefits. The cohabitation and marriage registries include data on flatmate and spouse pairs. Additionally, I have access to demographic data such as sex, age, nationality, country of birth, neighborhood, residency, and death. I also link information on parents and siblings for balancing checks.

Survey Data.—I merge two survey datasets: National Student Survey (NSS) and Longitudinal Internet Studies for the Social Sciences (LISS). The NSS panel includes questions on attitudes towards internationalization (Landelijk Centrum Studiekeuze 2024). The survey has been running since 2010 and contains aggregated responses from first-year students by program, university, and year of enrollment. These survey data are linked to administrative data using identifiers based on program, university, and year of enrollment.

The LISS panel includes questions on attitudes towards migration. The LISS panel is a nationally representative annual household survey that has been running since 2007 (CentERdata 2024). The survey contains individual responses to a wide range of topics related to economic, social, and health outcomes. It consists of around 5,000 households, comprising approximately 7,500 individuals, drawn from a random sample of Dutch households. Since the LISS panel is representative of the full Dutch population, only a subset of survey respondents can be matched to the sample of university students. Approximately 15 percent of the Dutch population holds a university degree, which corresponds to the share of LISS respondents that could be merged with the administrative data of university students. The merging is done using the same unique identifier that links all registries. The average age of respondents is 30.7.

Sample Construction.—I focus on full-time students enrolled in bachelor's programs between 1988 and 2019. From the international student sample, I exclude exchange students (0.5 percent of the sample) due to their short period of study. From the native student sample, I exclude first-generation immigrants (5.5 percent) due to their uncertain status (whether they should be considered as native or

international) and students older than 30 at the time of enrollment (1.5 percent) as they do not represent the usual undergraduate populations. The results remain robust to the inclusion of first-generation migrants and students older than 30, as reported in the Supplemental Appendix.

I define a native student as someone who holds Dutch nationality. To define an international student, I apply two criteria: not having Dutch nationality at the time of enrollment and not having completed secondary education in the Netherlands. This definition aims to capture individuals who lack prior exposure to the Dutch education system. I discuss the consequences of using alternative definitions of international students in the Robustness section.

The analysis focuses on the first enrollment of students in bachelor's programs. This approach ensures accurate assignment of the treatment variable, defined as the share of international students at the time of a student's first enrollment. For instance, if a student enrolls in a program, subsequently drops out, and later enrolls in a different program, they are included in the study only once.

One limitation of the data is the lack of information on tutorial group assignments within programs. While students attend lectures together, tutorials are held in smaller groups. This allocation is handled centrally and typically does not allow for student choice. Depending on administrative logistics, students may remain with the same group across all courses or be reassigned for each course or block. Importantly, there is generally no input from students, parents, or faculty. As such, there is little scope for selection into classrooms within a program.

B. Identification Strategy

The aim of this paper is to estimate the impact of exposure to international students on the long-term outcomes of native students. Building on the model specifications of Carrell, Hoekstra and Kuka (2018) and Anelli, Shih and Williams (2023), I exploit idiosyncratic variation in the share of international students within a bachelor's program over time, using the following model:

$$(1) \quad Y_{ifut} = \beta D_{fut} + \alpha_{fu} + \tau_{ft} + \varepsilon_{ifut},$$

where Y_{ifut} measures an outcome of interest⁷ of a native student i in a field of study f at a university u enrolled in a year t ; $D_{fut} = \frac{Int_{fut}}{N_{fut} - 1}$ is the share of international students; α_{fu} are the *field of study* $\times$ *university* fixed effects; τ_{ft} are the *field of study* $\times$ *year of enrollment* fixed effects. Field of study refers to disciplines such as economics, business administration, sociology, etc. In total there are 199 fields of study, 13 universities, and 32 years. For brevity I refer to α_{fu} as program fixed effects and to τ_{ft} as cohort fixed effects. I cluster standard errors at the intersection of field of study and university level as the treatment is assigned at this level (Abadie, Athey, Imbens, and Wooldridge 2023). Additionally, I report q -values using the method proposed by Anderson (2008) to account for multiple hypothesis testing.

⁷ Supplemental Appendix A provides a detailed description of the variable construction.

The inclusion of program fixed effects accounts for time-invariant characteristics, such as the location of the program, which could influence student selection across programs. Cohort fixed effects control for time-varying factors that affect all students, such as changes in tuition fees. I check the robustness of the main findings against alternative model specifications that include linear and quadratic field specific time trends, program size, and individual and peer characteristics. The results are robust to the inclusion of these controls and are reported in the Supplemental Appendix.

C. Identifying Variation

By leveraging the institutional setting of Dutch and EU law and incorporating program and cohort fixed effects, the primary source of variation in Dutch students' exposure to international peers arises from differences in cohort entry, which are largely driven by birth years. This variation arises from fluctuations in both the number of native students and the number of international students within programs.

First, the number of native students varies due to natural fluctuations in Dutch birth cohorts, leading to random changes in university enrollment sizes. Second, the number of international students fluctuates for two reasons. Admission offices at Dutch universities have no control over the number of both Dutch and EEA+ students, resulting in some randomness influenced by population changes in other European countries. Additionally, various push factors from EEA+ and non-EEA+ countries further contribute to these fluctuations. In summary, the variation in the number of both native and international students entering Dutch universities is largely driven by natural cohort fluctuations, external demographic changes, and push factors from other countries, creating a degree of randomness in student inflow.

As discussed in Angrist (2014), there needs to be sufficient variation in the cohort-to-cohort changes of the share of international students. Figures A1 and A2 in the Supplemental Appendix illustrate the variation in the share of international students, both before and after accounting for the fixed effects.⁸ Even after controlling for program and cohort fixed effects, there is still enough variation to precisely estimate the parameter of interest. Specifically, one standard deviation of the share of international students is 12.6 percentage points, while the residualized share has a standard deviation of 5.9 percentage points. To put this in perspective, an increase of one standard deviation in the residualized share translates to adding approximately eight international students to an average program of 137 students. This year-to-year variation is largely due to Dutch universities having no control over the number of incoming Dutch and EEA+ nationals due to the Dutch and EU laws, leading to idiosyncratic fluctuations in the share of international students.

⁸There remains meaningful residual variation in the samples merged with the NSS and LISS panels: The standard deviation of the residualized share is 3.9 percentage points in the NSS sample and 3.2 percentage points in the LISS sample (see Supplemental Appendix Figures A3 and A4 for the NSS sample). Due to privacy regulations, histograms for the LISS sample cannot be plotted, as the cell sizes are small.

D. Identification Challenges

In this section I discuss three challenges to my identification strategy: selection, reflection, and common shocks.

Selection.—My identification strategy does not rule out student selection *across* programs. For example, if students know from previous years that there are many international students in economics at Maastricht University due to its proximity to the border with Germany and Belgium and fewer international students in economics at Erasmus University, and they decide where to enroll based on this knowledge, then this is not a violation of my identification strategy. This is because I do not compare students across different programs (economics at Maastricht University versus economics at Erasmus University). Instead, I compare students *within* the same program *across* different cohorts (economics at Maastricht University in 2010 versus economics at Maastricht University in 2011).

The identification strategy relies on the assumption that students do not select into cohorts *within* a program based on the presence of international students. For example, if students decide to enroll in Economics at Maastricht University in 2011 and not in 2010 because of the different share of international students, then this violates my identification strategy. This is unlikely due to two reasons. First, prospective applicants do not observe the composition of students in the year they enroll until their first day in the classroom. Only when students enroll in a program and start their first day can they observe this composition. That is why I measure the share of international students and all other peer outcomes at the start of the academic year. For example, if some students enroll after the start of the academic year, I do not include them in my sample, and they are not included to measure peer characteristics.

Second, even if students could anticipate the approximate composition, they would have to react to that information to violate my identification strategy. In other words, if students delay (enroll next year instead of this year) or accelerate (enroll this year instead of next year) their enrollment into a specific program based on the share of international students in this program, then it would violate this assumption. There are two cases of how students can do that. Imagine that students can observe the approximate composition of their peers and decide to either delay or accelerate their enrollment. To delay enrollment, students who would not have otherwise taken a gap year choose to take one. Conversely, to accelerate enrollment, students who would have otherwise taken a gap year decide to skip it.

To examine whether native students react to the share of international students, I look at whether the probability of taking a gap year changes, as well as a battery of other observable characteristics. I test for selection using balancing tests, regressing observable characteristics on native students' exposure to international students, using the following model:

$$(2) \quad X_{ifut} = \beta D_{fut} + \alpha_{fu} + \tau_{ft} + \varepsilon_{ifut},$$

where X_{ifut} includes student, sibling, and parent characteristics.

Supplemental Appendix Table A1 reports estimates with and without program and cohort fixed effects. Without fixed effects, many estimates are statistically significant. With fixed effects, only 1 of 32 estimates remains significant at the 1 percent level. I do not reject the null hypothesis that the coefficients are jointly equal to zero. Consistent with the scenario discussed above, students are neither more nor less likely to take gap years in response to the presence of international students. It is worth noting that the lack of statistical significance is not due to large standard errors but rather due to the small size of both the coefficients and the standard errors.

To assess the possibility of serial selection across cohorts, where students base their enrollment decisions on the international student composition of previous cohorts, I test whether the share of international students in year $t - 1$ predicts the observable characteristics of Dutch students in year t . Supplemental Appendix Table A2 shows that only 4 of the 32 estimates is statistically significant at the 10 percent level. I do not reject the null hypothesis that the coefficients are jointly equal to zero.

Further, I regress the characteristics of peers of native students in a program on the share of international students in that program, using the following model:

$$(3) \quad \bar{X}_{fut} = \beta D_{fut} + \alpha_{fu} + \tau_{ft} + \varepsilon_{fut},$$

where $\bar{X}_{fut}$ includes student, sibling, and parent characteristics collapsed to the program level. Supplemental Appendix Table A3 shows that only 1 of 32 estimates is significant at the 10 percent level. Again, I do not reject the null hypothesis that the coefficients are jointly equal to zero.

Supplemental Appendix Table A4 examines the correlation between the share of international students and indicators of program quality. As proxies for program quality, I use the program size and the outcomes of graduates from the same year the share of international students is measured. I regress the outcomes of graduates from a given program and graduation year on the share of international students to whom first-year native students in that same program and year are exposed. For example, I regress outcomes of students who graduated from economics at Maastricht University in 2010, such as their employment five years after graduation, on the share of international students among first-year economics students at Maastricht University in 2010. The idea is that the outcomes of graduates reflect the quality of the program.

With fixed effects, none of the 18 indicators of program quality is statistically significant at conventional levels, and the coefficients are precisely estimated. These findings suggest that, conditional on program and cohort fixed effects, the share of international students does not correlate with changes in the overall quality of programs. The lack of significant differences supports the assumption that any observed effects of international students on native peers' outcomes are not driven by systematic variations in program quality.

In total, out of 114 estimates, one would expect to have 1 percent of coefficients significant at the 1 percent level, 5 percent at the 5 percent level, and 10 percent at the 10 percent level. The share of significant estimates in my sample is at or below the respective expectations: they are 1 percent, 1 percent, and 5 percent, respectively.

TABLE 1—THE EFFECT OF EXPOSURE TO INTERNATIONAL STUDENTS ON SOCIAL TIES AND MIGRATION DECISIONS OF NATIVE STUDENTS 15 YEARS POST-ENROLLMENT

	Cohabited with a nonnative (1)	Married to a nonnative (2)	Emigrated (3)
International share in 10%-points	0.006 (0.001)	0.001 (0.001)	0.003 (0.001)
Sharpened <i>q</i> -value	[0.005]	[0.042]	[0.019]
Mean	0.102	0.035	0.065
Effect size	5.9%	4.2%	4.0%
Cohorts	1988–2008	1988–2008	1988–2008
Observations	605,367	587,676	617,094

Notes: All regressions include program and cohort fixed effects. A nonnative is defined as a person without Dutch nationality. Effect size represents the ratio of the estimated coefficient to the sample mean of the outcome. Sharpened *q*-values are adjusted False Discovery Rate *p*-values, calculated using the method proposed by Anderson (2008), to account for multiple hypothesis testing. Standard errors, which are clustered at the program level, are reported in parentheses.

Reflection and Common Shocks.—Apart from the selection problem, identifying peer effects is complicated by two more issues: the reflection and common shocks problems, as outlined by Manski (1993). The reflection problem arises when international and native students potentially influence each other’s outcomes simultaneously, while common shocks occur when factors such as curricular revisions affect the entire group. These issues emerge when contemporaneous outcomes are used as treatment variables. However, in my setting these problems are mitigated by using the share of international students as the measure of peer influence. This variable is fixed at the start of the program and is unaffected by other variables or common shocks.

III. Results

In this section I present the main findings on the impact of exposure to international students on social ties, migration decisions, attitudes towards internationalization and migration, and labor market outcomes. Next, I conduct a heterogeneity analysis based on sex, field of study, share of females, program size, and the origin of international students. Finally, I conclude with robustness checks and a meta-analysis.

A. Main Results

Effect of Exposure on Social Ties.—I estimate the impact of studying with international students on the likelihood of living together and forming partnerships with nonnative individuals 15 years after enrolling in a program. All coefficients represent a 10 percentage point increase in the share of international students in the first year of a bachelor’s program.⁹ Column 1 of Table 1 shows that a 10 percentage point increase

⁹Supplemental Appendix Figures A5 and A6 plot binned relationships between the residualized share of international students and the residualized outcomes.

TABLE 2—THE EFFECT OF EXPOSURE TO INTERNATIONAL STUDENTS ON THE SOCIAL TIES OF NATIVE STUDENTS OUTSIDE THE PROGRAM

	Cohabited with a nonnative outside the program university		Married to a nonnative outside the program university	
	(1)	(2)	(3)	(4)
International share in 10%-points	0.006 (0.001)	0.005 (0.001)	0.001 (0.001)	0.001 (0.001)
Sharpened <i>q</i> -value	[0.005]	[0.005]	[0.042]	[0.044]
Mean	0.101	0.100	0.035	0.035
Effect size	5.5%	4.9%	3.9%	3.8%
Cohorts	1988–2008	1988–2008	1988–2008	1988–2008
Observations	605,367	605,367	587,676	587,676

Notes: All regressions include program and cohort fixed effects. A nonnative is defined as a person without Dutch nationality. Effect size represents the ratio of the estimated coefficient to the sample mean of the outcome. Sharpened *q*-values are adjusted False Discovery Rate *p*-values, calculated using the method proposed by Anderson (2008), to account for multiple hypothesis testing. Standard errors, which are clustered at the program level, are reported in parentheses.

in the share of international students increases the likelihood that native students will live with nonnatives by 0.6 percentage points 15 years post-enrollment. This change corresponds to a 5.9 percent increase relative to the sample mean of 10.2 percent.

Native students are also more likely to marry a nonnative after exposure to more international students at university. A 10 percentage point increase in the share of international students leads to a 0.1 percentage points increase in marriages between natives and nonnatives (see column 2 of Table 1). This translates to a 4.2 percent increase of the sample mean of 3.5 percent. These findings illustrate how interactions with international students can foster closer social connections between natives and nonnatives, contributing to a more integrated and culturally diverse society.

One explanation for the increased likelihood of cohabiting and marrying nonnatives due to exposure to international students is the larger pool of nonnatives. More international students in a program increase the chance of cohabitation and marriage between native and international students from the same program. Table 2 shows that native students are more likely to cohabit and marry nonnatives outside their program and university. These findings suggest the increased likelihood of such relationships reflects deliberate choices rather than merely convenient meeting opportunities.

Effect of Exposure on Migration Decisions.—Studying alongside international students can broaden native students' perspectives and increase their interest in other countries. For example, it can affect the propensity of native students to seek experiences abroad. Column 3 of Table 1 shows that a 10 percentage point increase in the share of international students increases the probability of emigration by 0.3 percentage points, which is a 4 percent increase of the sample mean of 6.5 percent. This suggests that universities make native students more internationally oriented.

TABLE 3—THE EFFECT OF EXPOSURE TO INTERNATIONAL STUDENTS ON ATTITUDES TOWARDS INTERNATIONALIZATION AND MIGRATION OF NATIVE STUDENTS

	Coefficient	SE	Sharpened <i>q</i> -value	Mean	Observations
<i>Panel A. Attitudes towards internationalization</i>					
1. How satisfied are you with the level of internationalization? (1–very dissatisfied, 5–very satisfied)	0.053	(0.019)	[0.015]	3.524	123,038
2. How satisfied are you with the level of encouragement to learn about other cultures? (1–very dissatisfied, 5–very satisfied)	0.038	(0.019)	[0.042]	3.197	123,038
<i>Panel B. Attitudes towards migration</i>					
1. Do you agree that people of foreign origin who legally reside in the Netherlands should be entitled to the same social security as Dutch citizens? (1–fully disagree, 5–fully agree)	0.238	(0.095)	[0.021]	3.790	1,180
2. Do you agree that it does not help a neighborhood if many people of foreign origin or descent move in? (1–fully disagree, 5–fully agree)	–0.138	(0.062)	[0.031]	3.258	1,180
3. Where would you place yourself on a scale from 1 to 5, where 1 means that European unification has already gone too far and 5 means that it should go further?	0.239	(0.068)	[0.005]	2.148	1,166

Notes: All regressions include program and cohort fixed effects. A coefficient represents a 10 percentage point increase in the share of international students. Sharpened *q*-values are adjusted False Discovery Rate *p*-values, calculated using the method proposed by Anderson (2008), to account for multiple hypothesis testing. In panel A outcomes are based on National Student Survey data. In panel B outcomes are based on Longitudinal Internet Studies for the Social Sciences Survey data. Standard errors, which are clustered at the program level, are reported in parentheses.

Effect of Exposure on Attitudes towards Internationalization and Migration.—Exposure to international students might also shift native students' preferences, making them more open to other cultures. To examine this, I analyze how international student exposure influences natives' attitudes towards internationalization and migration using two survey datasets.

I find that exposure to international students positively affects students' perceptions of internationalization (see panel A of Table 3).¹⁰ A 10 percentage point increase in the share of international students increases satisfaction with internationalization by 0.053 points (0.12 SD) and satisfaction with encouragement to learn about other cultures by 0.038 points (0.07 SD).

Exposure affects views of natives not only during university but also post-graduation (see panel B of Table 3).¹¹ A 10 percentage point increase in the share of international students in a university program leads to a 0.238 point increase (0.30 SD) in

¹⁰In panel A of Supplemental Appendix Table A5, I check for selection bias in programs that participate in the NSS panel and their responses to internationalization-related questions. I find no correlation between the share of international students and the probability of a program being in the NSS panel or answering any of the internationalization-related questions. Panels B and C of Supplemental Appendix Table A5 show the main findings using linear regression with weights and ordered probit regression. The results are consistent with those from the linear regression in Table 3.

¹¹In panel A of Supplemental Appendix Table A6, I check for selection bias of native students into the LISS panel and their responses to immigration-related questions. I find no correlation between the share of international students and the probability of being in the LISS panel or answering any of the immigration-related questions. Panels B and C

TABLE 4—THE EFFECT OF EXPOSURE TO INTERNATIONAL STUDENTS ON LABOR MARKET OUTCOMES OF NATIVE STUDENTS 15 YEARS POST-ENROLLMENT

	Employed (1)	Income percentile (2)	Entrepreneur (3)	% of foreign-born coworkers (4)
International share in 10%-points	−0.001 (0.001)	0.001 (0.001)	0.000 (0.001)	0.001 (0.001)
Sharpened <i>q</i> -value	[0.077]	[0.191]	[0.257]	[0.109]
Mean	0.943	0.717	0.072	0.122
Effect size	−0.1%	0.1%	0.1%	0.6%
Cohorts	1988–2008	1988–2008	1988–2008	1991–2008
Observations	576,961	576,961	576,961	427,687

Notes: All regressions include program and cohort fixed effects. Effect size represents the ratio of the estimated coefficient to the sample mean of the outcome. Sharpened *q*-values are adjusted False Discovery Rate *p*-values, calculated using the method proposed by Anderson (2008), to account for multiple hypothesis testing. Standard errors, which are clustered at the program level, are reported in parentheses.

agreement that people of foreign origin who legally reside in the Netherlands should be entitled to the same social security as Dutch citizens. Furthermore, it leads to a 0.138 point decrease (−0.16 SD) in agreement that it does not help a neighborhood if many people of foreign origin or descent move in. Additionally, exposure to international students shifts opinions on European unification, with a 0.239 point increase (0.28 SD) in the belief that European unification should go further.

The results indicate that exposure to international students leads to deliberate and lasting social ties between native and nonnative students. Native students are more likely to form relationships with nonnatives outside their academic environment, reflecting a preference shift. This exposure also influences migration decisions and shapes natives' views on internationalization and migration, highlighting the important role universities play in encouraging long-term social integration and openness.

Effect of Exposure on Labor Market Outcomes.—After establishing the positive effects of exposure on the social ties and openness to migration among native students, I examine whether the presence of international students impacts the labor market outcomes of natives. Policymakers often express concerns that international students might strain educational resources and negatively affect the quality of education, potentially disadvantaging native students' labor market outcomes. Table 4 presents the findings for employment, income percentile, entrepreneurship, and the share of foreign-born coworkers. Across all outcomes, the estimates are precisely estimated zero. The precision of the estimated coefficients allows me to rule out any effects more negative than −0.2 percentage points on employment, income, and entrepreneurship, as well as effects more negative than −0.1 percentage point on the

of Supplemental Appendix Table A6 present the main findings using linear regression with weights and ordered probit regression. The results are similar to those from linear regression without weights in Table 3.

share of foreign-born coworkers.¹² Overall, the findings point to a lack of long-term impact on the labor market.

This absence of negative effects on labor market outcomes is particularly significant in light of ongoing policy debates that often focus on the potentially detrimental impacts of international student presence on the educational and labor market outcomes of natives.¹³ The precision of these estimates refutes concerns about statistically and economically significant negative effects, which suggests that exposure to international peers does not harm native students' local economic success.

B. Nonlinearity

The effects of exposure to international students may plausibly be nonlinear. For instance, Anderberg et al. (2024) document an inverse-U shaped relationship between immigrant share and natives' in-group bias: Bias initially increases with diversity but declines once immigrants form a sufficiently large group to reshape the peer environment. Motivated by this insight, I examine whether the relationship between the share of international students and native students' outcomes similarly deviates from linearity. Following the approach of Anderberg et al. (2024), I estimate a cubic specification of the form

$$(4) \quad Y_{ifut} = \beta_1 D_{fut} + \beta_2 D_{fut}^2 + \beta_3 D_{fut}^3 + \alpha_{fu} + \tau_{ft} + \varepsilon_{ifut}.$$

After deriving the marginal effects, I use the delta method to compute standard errors and plot the results with 95 percent confidence intervals in Supplemental Appendix Figures A7 and A8.

Unlike Anderberg et al. (2024), I find limited evidence of nonlinear effects. The marginal effects are consistently positive for social outcomes, such as cohabitation and marriage, indicating that greater exposure to international peers strengthens

¹² Emigration is not a random event. In the period prior to leaving the Netherlands, Dutch students who later emigrated were less likely to be employed, ranked lower in the income distribution, were less likely to be entrepreneurs, and tended to work in more international firms (Supplemental Appendix Table A7). These patterns suggest that emigrants were less attached to the domestic labor market. Since realized outcomes are not observed for emigrants, I estimate effects of exposure only for those who remain in the country and address missing outcomes through bounding analyses. In the first approach, I impose extreme assumptions by assigning emigrants either very low or very high outcomes. For binary outcomes (employment and entrepreneurship), I impute zeros for lower bounds and ones for upper bounds. For continuous outcomes (income percentile and the share of foreign-born co-workers), I use values from the bottom 10 percent and top 10 percent of peers in the program of emigrants. Panels A and B of Supplemental Appendix Table A8 show that the resulting bounds are tight and close to zero. The only statistically significant estimate is the lower bound for employment, which suggests that exposure to a higher share of international students reduces employment by 0.3 percentage points, a small effect relative to the sample mean of 88.2 percent. In the second approach, I make a less extreme assumption and impute outcomes of emigrants from the year before their last observed year. The results, shown in panel C of Supplemental Appendix Table A8, are very similar to those from the upper- and lower-bound analysis, again producing tight bounds close to zero. The only exception is employment, with an estimated effect of -0.2 percentage points. Overall, the bounding analyses indicate that any bias from missing outcomes of emigrants is negligible, with estimated effects remaining very small and close to zero.

¹³ Supplemental Appendix Table A9 and panel A of Supplemental Appendix Table A10 report additional estimates of the effect of exposure to international students on educational outcomes of natives, including the probability of dropping out in the first year, switching programs within the same university, transferring to another university while remaining in the same program, time to graduation, and completion of bachelor's and master's degrees. In line with the labor market results, none of these outcomes are statistically significant.

native students' social ties across the entire distribution of exposure. For emigration, the marginal effects are significantly positive at exposure levels up to 20 percent, but becomes statistically insignificant at higher levels. This suggests that even modest exposure is sufficient to increase international mobility, while additional exposure does not further reinforce this effect. In contrast, the marginal effects on labor market outcomes are not statistically significant at any level of international student presence, with the exception of income. Overall, the findings indicate that the effects of exposure to international students are stable across the distribution of international student presence, with positive effects on social outcomes and no significant effects on labor market outcomes.

C. Heterogeneity

In this section I examine how international student exposure impacts groups with different social dynamics, competitive pressures, network diversity, and cultural similarities. These factors can influence both social integration and professional outcomes, with some environments encouraging stronger cross-cultural connections and others limiting them. By analyzing how exposure effects vary across these underlying features, I aim to identify which contexts enhance or weaken the benefits of international student exposure.

Sex.—Prior research suggests that men and women may experience social and professional networks differently (Beaman, Keleher, and Magruder 2018; Cullen and Perez-Truglia 2023). This could influence integration outcomes between native and international students by sex. To investigate this, I interact a female dummy with the share of international students. Panel A of Supplemental Appendix Table A11 indicates no statistically significant differences by sex, suggesting that male and female natives experience similar impacts from exposure to international students.

Field of Study.—Social engagement patterns can vary by field of study. STEM fields, with their competitive nature and intensive coursework, may provide fewer opportunities for social integration than non-STEM fields. Panel B of Supplemental Appendix Table A11 shows that non-STEM students benefit from social integration with international students, while STEM students show no significant increase in the probability to cohabit with a nonnative. This suggests that the competitive environment of STEM fields may limit opportunities for cross-cultural interactions.¹⁴

Share of Females.—Research suggests that men tend to be more competitive than women (Niederle and Vesterlund 2007; Buser, Niederle, and Oosterbeek 2014). In highly male-dominated settings, competition may take precedence over collaboration, potentially limiting opportunities for social integration. To test this, I

¹⁴ Consistent with Anelli, Shih, and Williams (2023), I find evidence of “native flight” from STEM programs. I find that natives in STEM programs are more likely to drop out when exposed to more international students (see panel B of Supplemental Appendix Table A10). This contrasts with non-STEM fields, where I find no such effect, in line with the broader pattern that positive social effects are concentrated outside STEM.

examine whether the effect of exposure to international students varies by the sex composition of a program. I use a 20 percent female threshold because programs with such a low share of women are heavily male dominated, where competitive norms are likely to be most pronounced. Panel C of Supplemental Appendix Table A11 shows that in programs with fewer than 20 percent female students, exposure to international students has no significant effect on the probability to cohabit with a nonnative. This finding suggests that the competitive nature in male-dominated environments may suppress the social integration benefits of international exposure.

Program Size.—Larger programs may promote greater social integration, as they offer broader networks and more opportunities for students to engage with peers. Panel D of Supplemental Appendix Table A11 confirms this, showing that natives in larger programs experience stronger effects on social ties and migration decisions. Exposure in larger programs also increases the likelihood of employment in firms with more foreign-born coworkers, suggesting that these broader environments foster integration and may encourage interest in global career paths.

EEA+ and Non-EEA+ Students.—Cultural and institutional similarities between EEA+ students and Dutch natives may facilitate social connections, as they often share cultural and institutional backgrounds. This familiarity likely lowers barriers to interaction, resulting in stronger integration outcomes. Columns 7 and 8 of Supplemental Appendix Tables A12 and A13 show that exposure to EEA+ students significantly boosts social integration. In contrast, exposure to non-EEA+ students, who may face greater cultural and linguistic differences, shows no significant integration effects. However, exposure to non-EEA+ peers increases satisfaction with opportunities to learn about other cultures, showing that these interactions promote cultural exchange. These results suggest that shared cultural contexts with EEA+ students enhance social ties, while interactions with non-EEA+ students contribute to broader cultural awareness, highlighting the value of diversity in the learning environment.

D. Robustness

The main analysis uses a model with program and cohort fixed effects. Here, I assess the robustness of the findings using alternative model specifications, sample definitions, treatment definitions, and a placebo test.

Correction for Multiple Hypothesis Testing.—Given the number of hypotheses tested, I apply the correction method from Anderson (2008) to control for multiple hypothesis testing. This correction reduces the risk of false positives, and the results remain statistically significant at the 5 percent level (see sharpened q -values in Tables 1–4).

Outcome Measurement at 10 and 25 Years.—In the baseline analysis, I measure outcomes 15 years post-enrollment. To test for effects over shorter and longer

horizons, I estimate outcomes at 10 and 25 years post-enrollment in Supplemental Appendix Table A14. Results confirm the baseline findings: increased social ties between natives and nonnatives, higher emigration rates, and no significant labor market effects.

Including Time Trends.—To control for potential confounding from time-related factors, I include linear and quadratic field specific time trends. Columns 2 and 3 in Supplemental Appendix Tables A15 and A16 show that the results remain robust to this adjustment.

Controlling for Program Size.—To ensure that the results are not influenced by the size of the programs, I include program size as a control variable in the model. Column 4 in Supplemental Appendix Tables A15 and A16 shows that the results remain robust.

Controlling for Individual Characteristics.—To account for possible confounding from individual characteristics correlated with the share of international students, I include individual controls. Column 5 in Supplemental Appendix Tables A15 and A16 shows that the results stay robust.

Controlling for Peer Characteristics.—Although individual characteristics do not impact estimates, the share of international students could be correlated with peer characteristics. In column 6 in Supplemental Appendix Tables A15 and A16, I control for these characteristics, and the results remain unchanged.

Including First-Generation Migrants.—The main analysis excludes first-generation migrants due to ambiguity in categorizing them as native or international. Column 8 in Supplemental Appendix Tables A15 and A16 shows that including them does not change the results.

Including Older First-Year Students.—To focus on typical undergraduate populations, I exclude students over 30 at enrollment in the main analysis. Column 9 in Supplemental Appendix Tables A15 and A16 shows that including them does not affect the results.

Sensitivity to Alternative Definitions of International Students.—Angrist (2014) and Feld and Zölitz (2017) highlight the potential for measurement error to bias estimates of peer effects. The availability of comprehensive administrative data in my research allows for an examination of how different definitions of international students influence the sensitivity of my findings. Various definitions have been used in the literature to identify international students, including:

- Having a different nationality (Borjas 2004, 2006; Shen 2016; Chevalier, Isphording, and Lisauskaite 2020; Anelli, Shih, and Williams 2023; Beine, Peri, and Raux 2023; Rakesh 2023; Chen 2023; Zhu 2023).
- Being born in a different country (Machin and Murphy 2017; Shih 2017).

- Residing in a different country before university enrollment (Costas-Fernández, Morando, and Holford 2023).
- Completing secondary education in a different country.

These variations in definitions pose two main challenges. First, they complicate the comparison of findings across studies, as it becomes unclear whether observed differences in peer effects are due to genuine variations or merely result from differing definitions. Secondly, definitions that inaccurately characterize international students can lead to biased estimates of peer effects.

To estimate the extent of potential biases, I propose a definition of a “true” international student as someone who is both a nonnational and did not complete secondary education in the host country. This definition aims to capture individuals who lack prior exposure to the host country’s education system.

Supplemental Appendix Tables A12 and A13 show that the results remain largely consistent across alternative definitions. On average, these definitions yield similar outcomes, suggesting that previous studies using different definitions produce unbiased results, and that differences in estimated peer effects across settings are due to genuine variations.

Placebo Analysis.—In this section I present the results of the placebo analysis. Instead of using the actual share of international students that native students are exposed to, I assign them a share of international students from the same program and year but a different university.¹⁵ For instance, a student enrolled in economics at the University of Amsterdam in 2010 is assigned the share of international students from economics at Erasmus University in 2010. This process is repeated 1,000 times, each time estimating the effect of this placebo treatment.

The resulting normal-looking distributions of simulated counterfactual effects confirm that the true estimates are located in the tails of the distributions, further validating the robustness of the findings. Supplemental Appendix Figures A9–A11 display plots with estimates from 1,000 simulated datasets, the mean estimated coefficients of the randomly assigned shares, and the actual estimated coefficients. The mean estimated effect of the randomly assigned shares is centered around zero for all outcomes. In contrast, the actual estimated coefficients lie in the tails of the distribution, providing further support for the validity of the identification assumption.

E. Discussion

To contextualize my estimates, I compare the findings of this study with two bodies of literature: (i) the spillover effects of international students on native peers and (ii) the effects of interacting with minority groups on prejudice.

Peer Effects of International Students.—I compare my estimates with the existing literature on the peer effects of international students. Figure 2 summarizes estimates from previous studies (Chevalier, Isphording, and Lissauskaite 2020; Anelli, Shih, and Williams 2023; Costas-Fernández, Morando, and Holford 2023; Rakesh 2023). Since

¹⁵ Programs that are only offered at one university in a given year are excluded from the estimation.

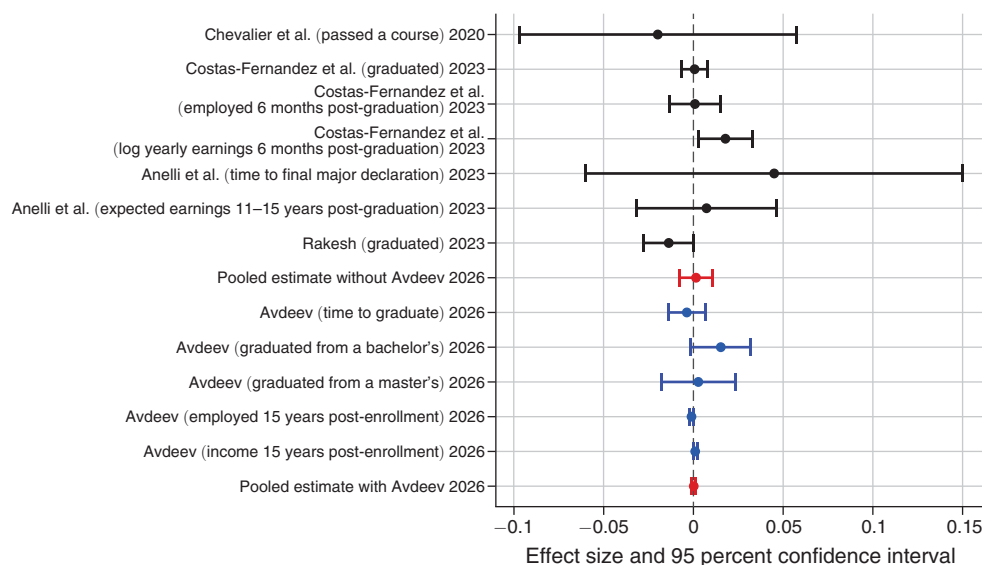


FIGURE 2. COMPARISON WITH ESTIMATES FROM THE INTERNATIONAL STUDENTS LITERATURE

Notes: This graph presents estimates from four studies (black) by Chevalier, Isphording, and Lissauskaite (2020); Costas-Fernández, Morando, and Holford (2023); Anelli, Shih, and Williams (2023); Rakesh (2023), two pooled random effects estimates calculated by the author (red), and estimates from the current paper (blue). One pooled estimate is based on the estimates from previous studies, while the second one additionally includes the estimates from the current paper. All estimates are converted to represent a 10 percentage point increase in the share of international students. Effect size represents the ratio of the estimated coefficient to the sample mean of the outcome. The estimate from Chevalier, Isphording, and Lissauskaite (2020) is reversed from “failed a course” to “passed a course” to ensure comparability. Supplemental Appendix Table A17 provides an overview of the estimates from previous research. Supplemental Appendix Table A9 shows the estimates of the current paper on educational attainment.

no meta-analytical study exists, I calculate a pooled random effects meta-analytic estimate for comparison.

While previous studies report small or no effects of international students on the educational outcomes of native students, they do not establish precise nulls. The pooled meta-analytic estimate, excluding my estimates, is 0.1 percent with a 95 percent confidence interval $[-0.009, 0.012]$. My estimates are more precise, allowing me to rule out even very small positive and negative effects. Moreover, I extend the analysis to long-term labor market outcomes, which have been previously unexplored. When my estimates are incorporated into the meta-analysis, the pooled estimate is 0.0 percent with a tighter 95 percent confidence interval $[-0.002, 0.002]$. This provides evidence that studying with international students has not only no statistically significant impact but also no economically meaningful effect on the educational and labor market outcomes of native students.

Interaction with Minority Groups on Prejudice.—I then compare my estimates on the effects of exposure to a higher share of international students on attitudes towards internationalization and migration with studies examining similar outcomes in Figure 3. For instance, Paluck, Green, and Green (2019) provides a meta-analysis of the effects of intergroup contact with various minority groups (ethnic, racial, etc.)

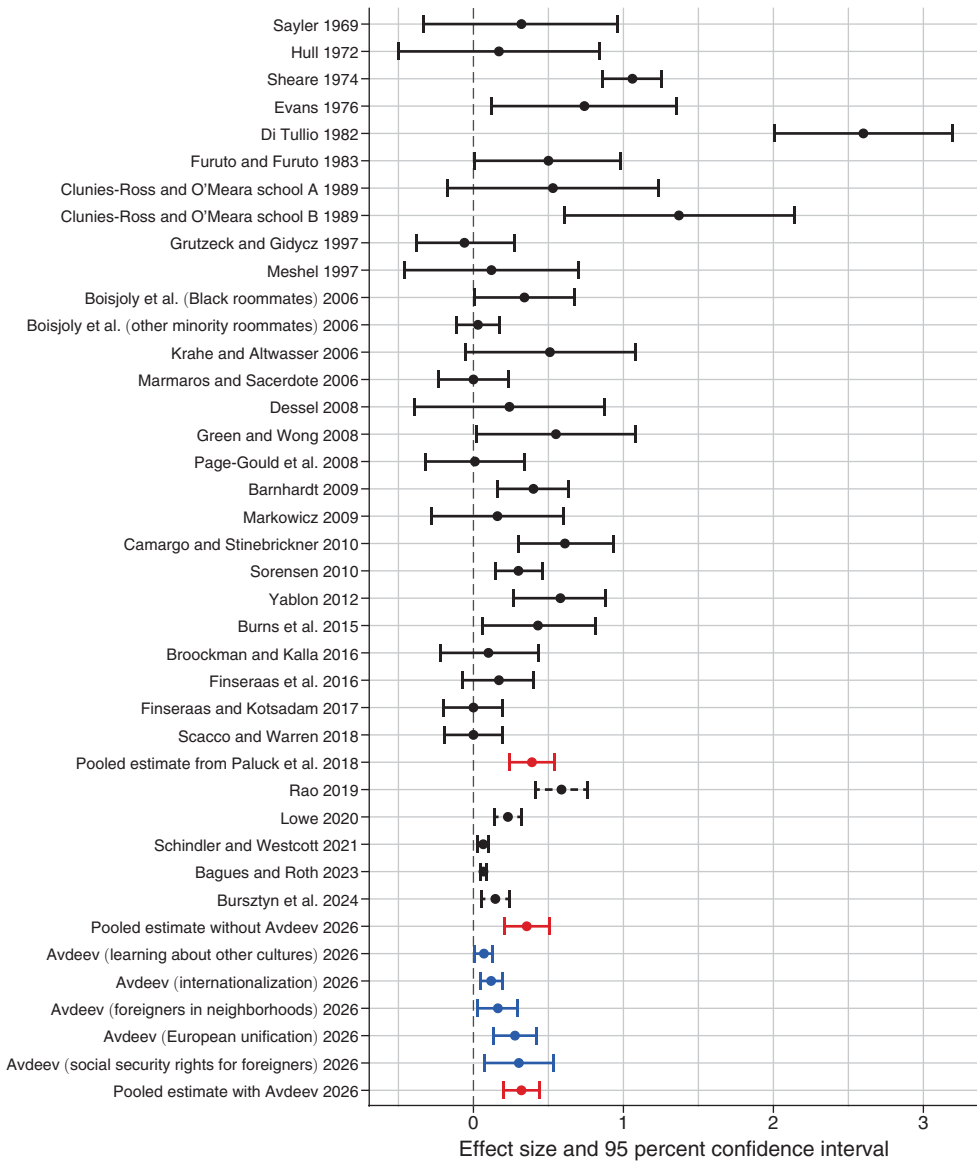


FIGURE 3. COMPARISON WITH ESTIMATES FROM THE CONTACT HYPOTHESIS LITERATURE

Notes: This graph presents estimates from 27 studies (black) included in a meta-analysis by Paluck, Green, and Green (2019); three pooled random effects estimates (red); estimates from Rao (2019); Lowe (2021); Schindler and Westcott (2021); Bagues and Roth (2023); and Bursztyn, Chaney, Hassan, and Rao (2024) (dashed); and estimates from the current paper (blue). One pooled estimate is from the meta-analysis by Paluck, Green, and Green (2019); while the second one additionally includes the estimates from Rao (2019); Lowe (2021); Schindler and Westcott (2021); Bagues and Roth (2023); Bursztyn, Chaney, Hassan, and Rao (2024); and the third one additionally includes estimates from the current paper. Paluck, Green, and Green (2019) includes studies that randomly assigned intergroup contact and measured outcomes more than a day after treatment. Effect size represents the ratio of the estimated coefficient to the standard deviation of the outcome. The estimate from Schindler and Westcott (2021) and the estimate for the effect of exposure on attitudes towards foreigners in a neighborhood (foreigners in neighborhoods) are reversed so that a positive coefficient indicates an improved attitude to ensure comparability.

on prejudice of the majority group. Paluck, Green, and Green (2019) estimates a pooled effect size of 0.39 SD with a 95 percent confidence interval [0.23, 0.55].

My three estimates of the effect on attitudes towards migration are 0.16 SD, 0.28 SD, and 0.30 SD in absolute terms. The latter two fall within the confidence interval of the pooled effect size from Paluck, Green, and Green (2019), suggesting that exposure to international students has a comparable impact to interventions studied in the contact hypothesis literature. This finding is notable because, unlike direct interventions, such as structured intergroup programs or assignment of roommates from different backgrounds, exposure to international students occurs more passively through everyday interactions. The fact that even this less intensive form of contact produces similar effects suggests that diversity in regular social environments can meaningfully shape attitudes towards migration.

IV. Conclusion

This paper provides the first evidence on the long-term impact of international students on native students. Using three decades of Dutch survey and administrative data and exploiting variation in the share of international students within university programs over time, this study identifies several key findings. A higher share of international students in the first year of a bachelor's program strengthens native students' connections with nonnatives, improves attitudes towards migration and cultural learning, and increases openness to migration. Importantly, exposure has no negative effects on labor market outcomes.

This study contributes to the literature on higher education and the contact hypothesis by focusing on the understudied group of international students. The findings underscore the dual importance of social and human capital considerations in shaping higher education funding and immigration policies. A better understanding of these relationships can help educational leaders and policymakers make informed decisions that enhance educational outcomes while fostering a more inclusive society. Policies that restrict international student enrollment may, therefore, limit broader societal benefits beyond tuition revenue.

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